POLITECNICO DI MILANO · ONLINE LEARNING PROJECT · 2026

Adaptive Budget Allocation for Online Advertising

Learning bids from auction outcomes while managing a limited campaign budget.

Rounds 1 … T: one bid decision each

One budget B for the whole campaign

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THE DECISION PROBLEM

A profitable bid must leave room for later rounds

A higher bid can increase the chance of winning, but each win reduces the money available later. The bidder learns from outcomes while deciding how much of the campaign budget to risk now.

1 · BID
Choose a bid b. A win is worth v to the campaign.

→

2 · AUCTION
Win if b reaches m, the highest competing bid.

→

3 · OUTCOME
A win earns v − b and costs b (first price).

→

4 · BUDGET
Every cost shrinks the money left for later rounds.

$$r_t = (v - b_t) \cdot 1\{b_t \geq m_t\} \qquad c_t = b_t \cdot 1\{b_t \geq m_t\} \qquad \Sigma_t\, c_t \leq B$$

Budget-aware LP policies cap spending at B/T, or at remaining budget divided by remaining rounds. Primal-dual control uses B/T as its average-cost target. Separate affordability checks enforce the realized budget.

Model: R1, cells 2–4; R2, cells 3, 5, 7, 10, 12. With several campaigns, r and c sum over selected arms.

OVERVIEW

Results from the saved simulations

Setting	Baseline	Revised	Measure · comparator
Single campaign	39.6	11.4	Final regret · stochastic clairvoyant LP
Multiple campaigns	595.2	70.1	Final pseudo-regret · stochastic clairvoyant LP
Primal-dual, stochastic	0.249	0.121	Pseudo-regret per round · best fixed distribution in hindsight
Primal-dual, changing sequence	0.222	0.134	Same measure and comparator
Segmented market, CUSUM-UCB	533.3	344.9	Final gap · segment-local oracle

Mean outcomes on synthetic auction instances. Each comparison uses paired market paths. Rows use different benchmarks and scales, so values are not comparable across rows. Definitions: appendix A1.

Trials: 10 · 20 · 10 per environment · 30. Sources: B1/R1 cell 8, R2 cell 13, R3 cell 17, B4/R4 cell 17.

Simulation conditions and paired comparisons

Scenario	Rounds	Budget	Main trials	Feedback
1. Single campaign	5,000	125	10	Selected bid’s win outcome
2. Multiple campaigns	3,000	700	20	Semi-bandit, selected arms
3. Primal-dual	3,000	700	10 per environment	Full reward/cost vectors after acting
4. Segmented market	3,000	560	30	UCB: semi-bandit. Primal-dual: full feedback

SYNTHETIC DATA

Simulated auctions only. No real campaigns.

Within each comparison, baseline and revised policies face the same market paths and use corresponding action seeds. Feedback differs between methods in Scenario 4.

Sources: R1 cells 2, 8 · R2 cells 2, 3, 5, 13 · R3 cells 1, 3, 5, 12, 17 · R4 cells 1, 2, 4, 6, 15.

SCENARIO 1 · SINGLE CAMPAIGN

Single campaign: learning from infrequent wins

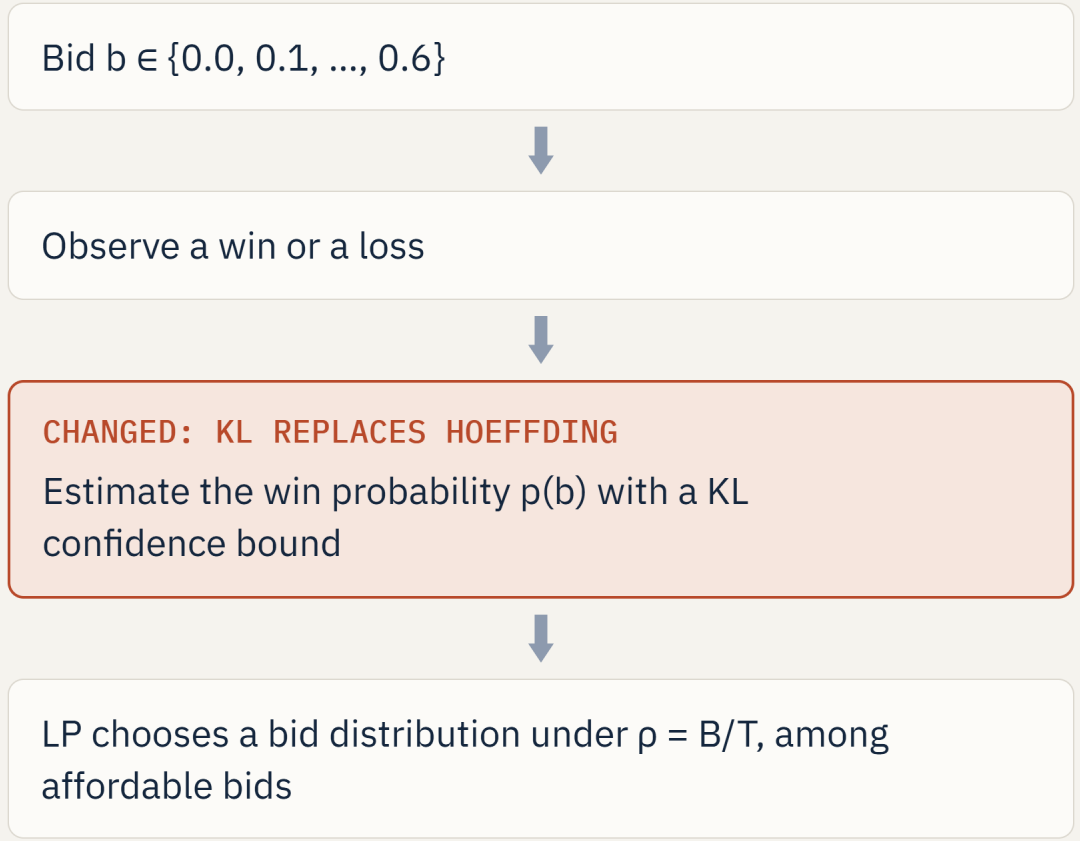
One campaign has value 0.6 per win and faces three independent uniform competitors. A bid b wins with probability b^3, so low bids produce few wins.

Baseline

Budget-blind UCB1 and budgeted Hoeffding UCB spend too early. The budgeted policy's loose cost lower bound can make the LP treat a bid as almost free.

Revision

KL bounds for the Bernoulli win probability, and bids above the remaining budget are excluded. The fixed cap $\rho = B/T$ stays.

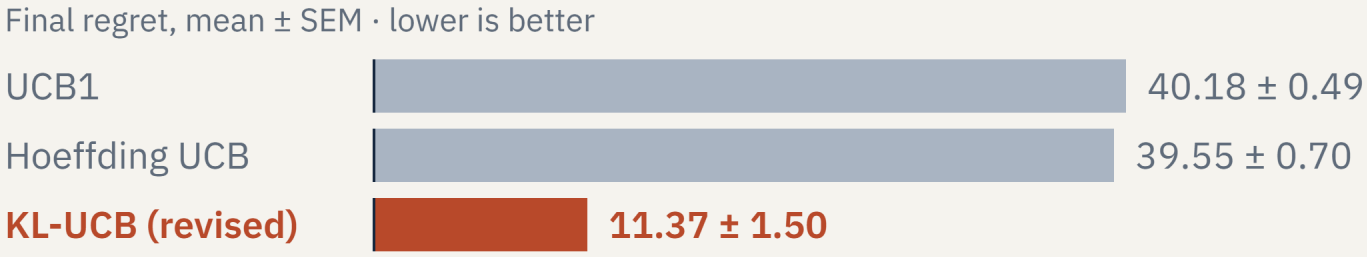


Sources: B1 cells 2, 3, 8 · R1 cells 2–4, 6. Horizon 5,000 rounds, budget 125.

SCENARIO 1 · RESULT

KL-UCB reduces single-campaign regret

Policy	Final regret, mean ± SEM	Mean spend / 125	Share of LP expected utility
Budget-blind UCB1	40.18 ± 0.49	124.29	36.4%
Budgeted Hoeffding UCB	39.55 ± 0.70	124.38	37.4%
Budgeted KL-UCB (revised)	11.37 ± 1.50	123.20	82.0%



The revised policy obtains more utility with similar total expenditure on these 10 paired trials.

Regret: expected stochastic-oracle utility minus realized learner utility.

10 paired trials. Sources: B1 cell 8; R1 cell 8 (UCB1, KL). Oracle LP: R1 cells 2, 5, 6.

SCENARIO 2 · MULTIPLE CAMPAIGNS

Multiple campaigns share one spending decision

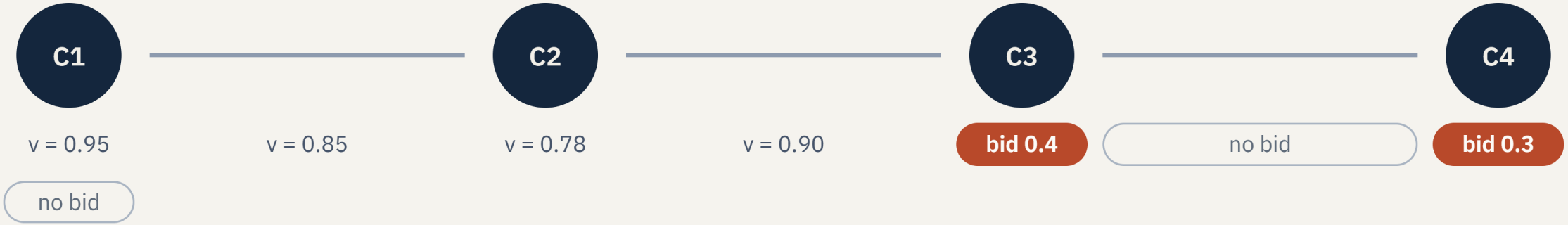
Four campaigns share a budget. Some campaigns conflict, so the bidder must choose a feasible combination of bids rather than optimize each campaign independently.

Baseline: Combinatorial-UCB spends too early.

Revision: keeps semi-bandit learning, with KL reward bounds, empirical expected costs and a remaining-budget cap.

$$\rho_t = \textit{remaining budget} / (T - t)$$

CONFLICT CHAIN · ADJACENT CAMPAIGNS CANNOT BID IN THE SAME ROUND



Illustrative feasible combination, not a recorded action

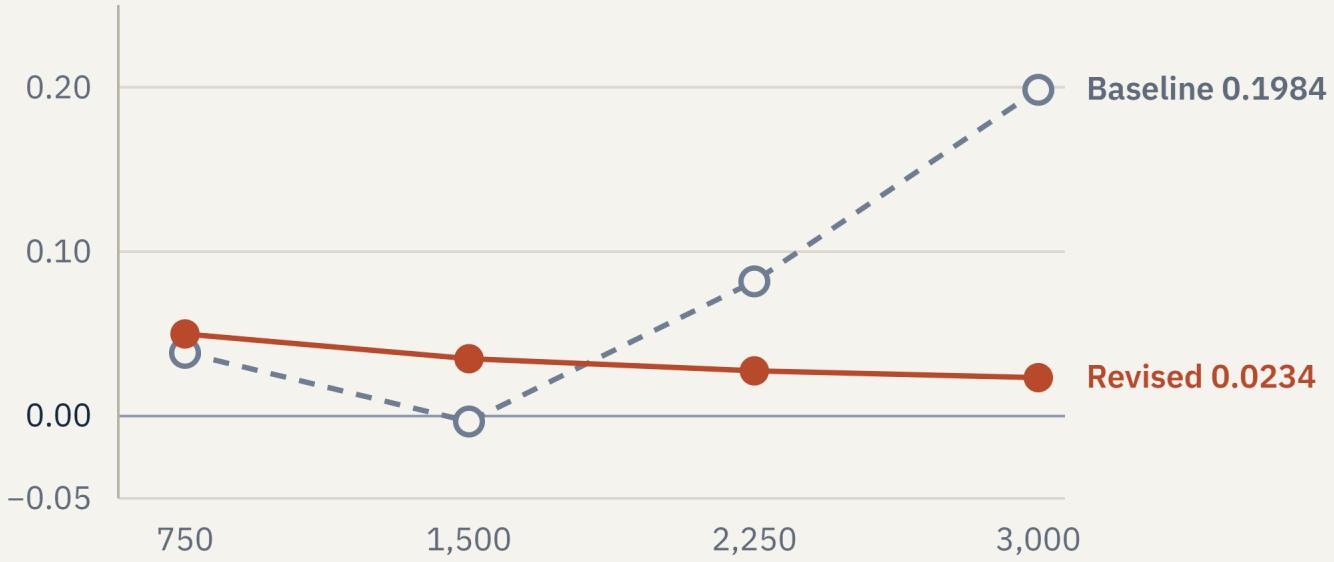
33 arms · 250 feasible combinations · B = 700 · T = 3,000

Sources: R2 cells 2, 3, 5, 7, 10, 12 · B2 cells 12, 14. t is counted from zero.

SCENARIO 2 · OVER THE HORIZON

Saved checkpoints show the effect of early spending

Pseudo-regret per round at four saved checkpoints · mean of 20 paired trials



Round. Markers are saved checkpoints; straight lines only guide the eye.

Round	Baseline	Revised
750	0.0385	0.0500
1,500	−0.0033	0.0349
2,250	0.0819	0.0275
3,000	0.1984	0.0234

The baseline's normalized pseudo-regret rises later in the campaign. The revised policy finishes with a smaller difference from the fixed stochastic oracle.

Sources: B2 cell 12, R2 cell 13 (saved checkpoint summaries). Comparator: true-mean LP, fixed distribution.

SCENARIO 2 · RESULT

The revised policy uses almost all the budget

70.05
final pseudo-regret, revised (baseline 595.15)

95.6%
of the stochastic LP's expected utility

699.84
mean spend of 700 (baseline 698.58)

Policy	Final pseudo-regret	Final realized regret	Mean utility	Mean spend / 700
Combinatorial-UCB baseline	595.15	596.32	1047.22	698.58
Revised KL-UCB	70.05	73.09	1570.45	699.84

Saved mean pseudo-regret 70.054544, or 70.1 at one decimal. Pseudo-regret is the main measure; the sampled clairvoyant policy's realized utility is not the LP objective.

20 paired trials. Source: R2 cell 13, corroborated by B2 cell 12. Oracle LP: R2 cell 7.

SCENARIO 3 · PRIMAL-DUAL CONTROL

Primal-dual control sets a price for spending

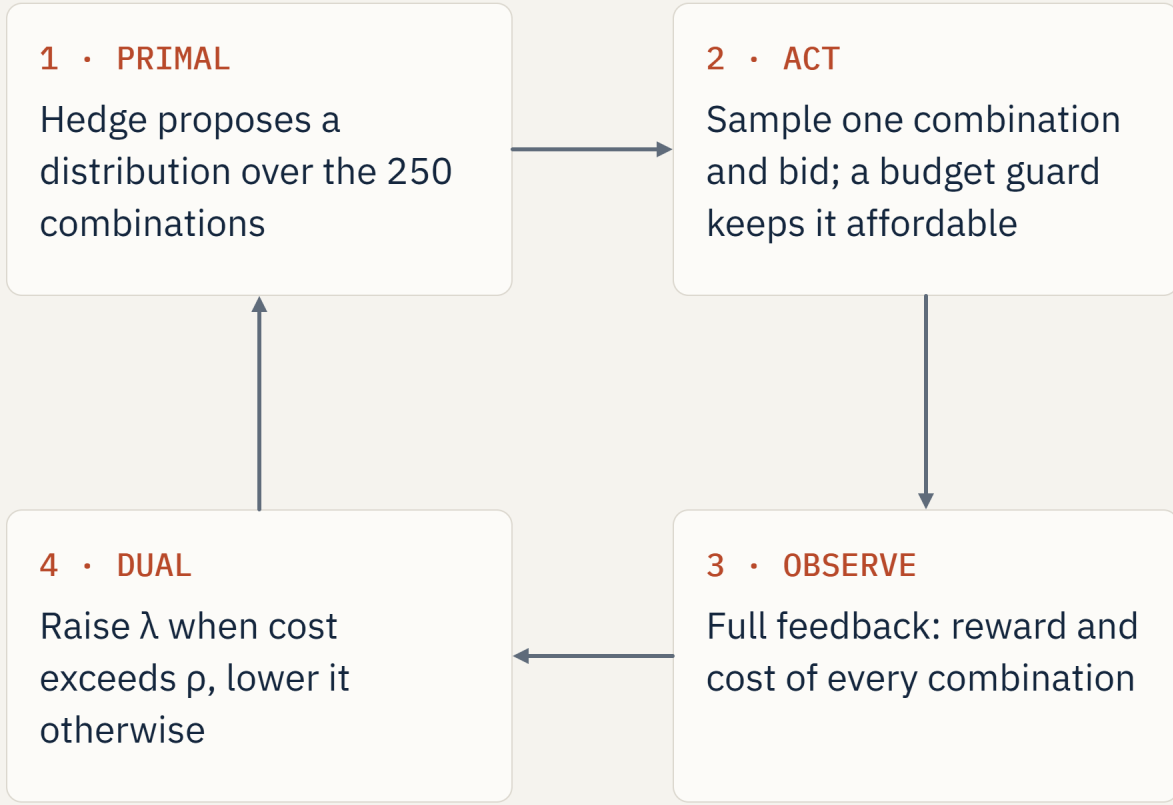
Full feedback makes every feasible combination's reward and cost available after acting. Hedge chooses a distribution over combinations; a dual multiplier λ raises the penalty for spending.

LAGRANGIAN SCORED BY HEDGE

$$L_t(a) = f_t(a) - \lambda_t\,(c_t(a) - \rho)$$

Baseline: conservative update rates leave budget unused.

Revision: Hedge and dual rates calibrated to their update ranges.



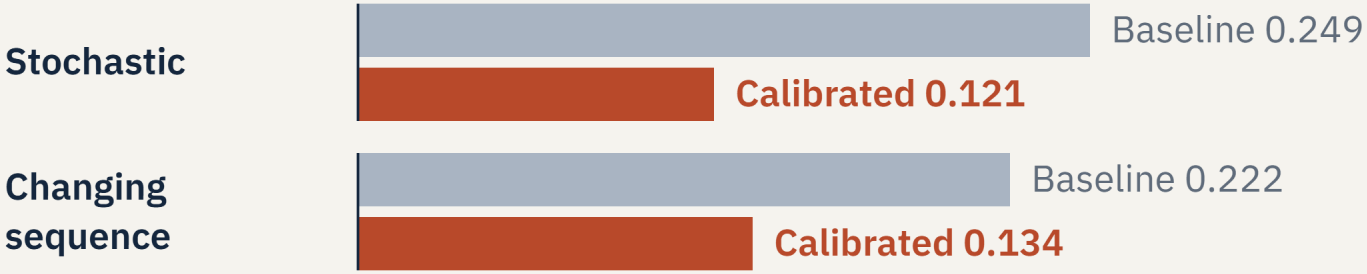
Sources: B3 cells 6, 8, 10 · R3 cells 1, 3, 5, 7, 8, 10, 12. Same instance as Scenario 2: T = 3,000, B = 700.

SCENARIO 3 · RESULT

Calibrated rates improve both tested environments

Environment	Version	Pseudo-regret / round	Realized utility	Mean spend / 700
Stochastic	Baseline	0.249	895.8	568.6
Stochastic	Calibrated	0.121	1287.9	675.8
Changing sequence	Baseline	0.222	835.4	557.6
Changing sequence	Calibrated	0.134	1100.0	596.8

Pseudo-regret per round · lower is better



10-trial paired comparisons against the best fixed distribution in hindsight. All main runs satisfy the hard budget.

Pseudo-regret uses proposed distributions; utility and spend use executed actions.

10 paired trials per environment. Sources: R3 cells 14, 17. Not comparable with Scenario 2 values.

SCENARIO 4 · CHANGING MARKETS

Changing markets require learning again

Competitor distributions change across three 1,000-round segments. A bid that was useful earlier may become expensive later, so information has a limited lifetime.

ROUNDS 1–1,000	CHANGE · ROUND 1,001	CHANGE · ROUND 2,001
Segment 1	**Segment 2**	**Segment 3**
Cheapest competition: Campaign 1	Cheapest competition: Campaign 2	Cheapest competition: Campaign 3

4 campaigns share budget 560 · 27 positive arms · 168 feasible combinations · 2 changes, known by construction

MAIN COMPARATOR · SEGMENT-LOCAL

One true-mean LP per segment, each with cap B/T

1,228.6 expected utility

STRONGER · SEGMENT-GLOBAL

Can redistribute expected spending across segments

1,252.0 expected utility

Sources: R4 cells 1, 2, 4, 6, 8. “Cheapest competition” = lowest Beta mean (cell 2). Oracles: expected values.

SCENARIO 4 · METHODS

Adaptation also needs budget for later segments

SEMI-BANDIT · UPDATING MARKET ESTIMATES

SW-UCB forgets

Uses only observations inside a calendar-round window. Revised window: 219 → 500 rounds.

CUSUM-UCB resets

Resets an arm's estimate after detecting a change. Detector settings unchanged.

CONTROLLING EXPENDITURE · SAME REVISED BUDGET LP

Empirical costs and remaining-budget pacing for both. Both baselines spend too early, limiting what they can learn later.

FULL FEEDBACK · SEPARATE

Primal-dual

Calibrated Hedge and dual rates.

Retains its full history: no forgetting.

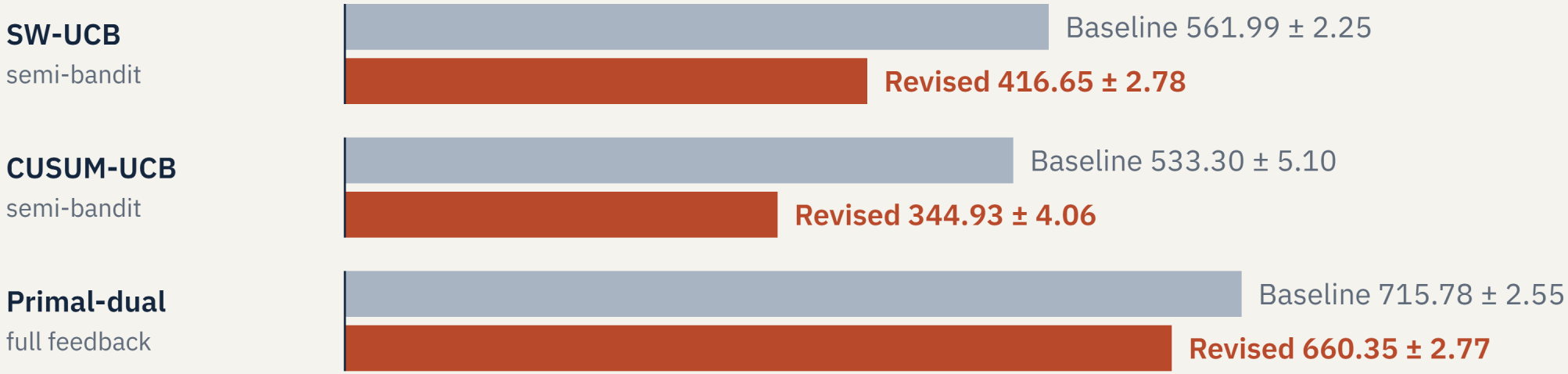
The baseline already has a per-action affordability check.

Sources: B4 cells 10, 12 · R4 cells 2, 10, 12, 15.

SCENARIO 4 · RESULT

CUSUM-UCB has the lowest recorded local-oracle gap

Final gap to the segment-local oracle · mean ± SEM over 30 paired trials · lower is better



Each gap subtracts realized learner utility from the same segment-local oracle's expected utility. The methods do not all receive the same feedback.

30 paired trials. Sources: B4 cell 17, R4 cell 17 (saved HTML tables). Not Scenario 3 pseudo-regret.

CONCLUSIONS

What the evidence supports

Supported finding	Evidence
KL bounds improve the saved sparse-win comparisons	Regret 11.37 vs 35.72 (Scenario 1); pseudo-regret 70.1 vs 100.5 (Scenario 2)
Empirical cost estimates improve the combinatorial cost ablation	Pseudo-regret 70.1 vs 166.2; spend by T/4 176.0 vs 241.4
Calibrated primal-dual rates improve both tested full-feedback environments	0.249 → 0.121 and 0.222 → 0.134 per round; the guard ablation shows no measured gain
A wider SW-UCB window improves the selected segmented instance	Local gap 496.1 with 219 rounds vs 416.6 with 500 rounds

LIMITS

Synthetic auction configurations and saved outputs only.

Joint changes do not identify every component’s contribution.

Finite horizon sweeps do not prove asymptotic behavior.

Feedback and benchmark differences constrain comparisons.

Ablation details in the appendix: A2 (Scenario 1), A3 (Scenario 2), A4–A5 (Scenario 3), A6 (Scenario 4).

Sources: R1 cells 16–17 · R2 cells 17–18 · R3 cells 17–21 · R4 cells 19–22 · docs/documentation-audit.md

REPOSITORY

Code, saved outputs and sources

github.com/antonio-arbizzani-tonelli/adaptive-budget-allocation-online-learning

Repository maintainer: Antonio Arbizzani Tonelli · linkedin.com/in/antonio-arbizzani-tonelli

Eight notebooks document the baselines and revisions. Scenario READMEs explain configurations, paired comparisons and limitations.

SCENARIO 1	SCENARIO 2	SCENARIO 3	SCENARIO 4
B1 · baseline_ucb	B2 · baseline_combinatorial_ucb	B3 · baseline_primal_dual	B4 · baseline_adaptive_learners
R1 · budget_aware_kl_ucb	R2 · dynamic_budget_pacing	R3 · calibrated_primal_dual	R4 · change_detection_and_pacing

Evidence reference: commit 38a050b32a6bd757c0c2c4db18c029caa039e919 · cell indices are zero-based and count markdown cells

Notebook guide: notebooks/README.md · Source map: docs/presentation-sources.md · Snapshot: docs/presentation-data.json

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APPENDIX

Technical backup for questions

A1 Benchmarks define what each difference means

A2 Scenario 1: confidence-bound ablation

A3 Scenario 2: confidence and cost ablations

A4 Scenario 3: guard ablation

A5 Scenario 3: pseudo-regret across tested horizons

A6 Scenario 4: window width and expenditure

APPENDIX A1 · MEASURES

Benchmarks define what each difference means

Scen.	Benchmark	Reported difference	Name
1	True-mean LP over a fixed bid distribution	Expected oracle utility – realized learner utility	Regret
2	True-mean LP over a fixed super-action distribution	Expected oracle utility – true expected utility of selected actions	Pseudo-regret
3	Best fixed distribution fitted to the complete realized sequence	Reward-vector difference between oracle and proposed learner distributions	Pseudo-regret
4	One true-mean LP per stationary segment, each capped at B/T	Expected oracle utility – realized learner utility	Gap, segment-local

Scenario 2 also stores realized regret and a sampled clairvoyant policy.

Scenario 4 also reports a stronger segment-global oracle that can move expected spending between segments.

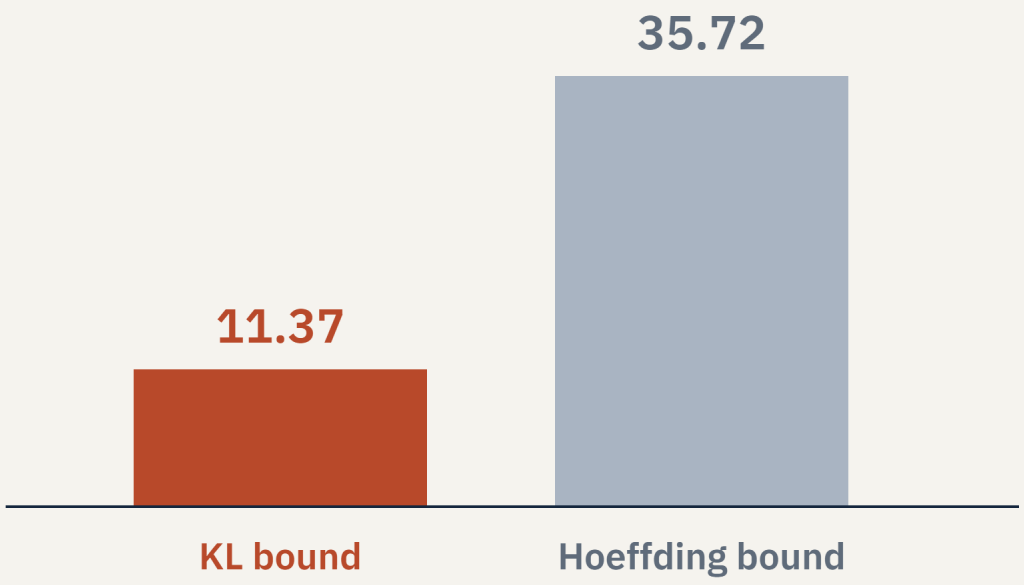
Sources: R1 cells 4, 6 · R2 cells 7, 12 · R3 cells 7, 12 · R4 cells 8, 17.

APPENDIX A2 · SCENARIO 1 · ABLATION

The confidence-bound ablation isolates one decision

The ablation keeps the revised policy structure and changes only the confidence bound.

Mean final regret · lower is better



Bound	Mean final regret	Mean spend / 125
KL	11.37	123.20
Hoeffding	35.72	45.55

Wider intervals can also keep the policy choosing low bids that rarely win. The ablation supports KL bounds on this instance.

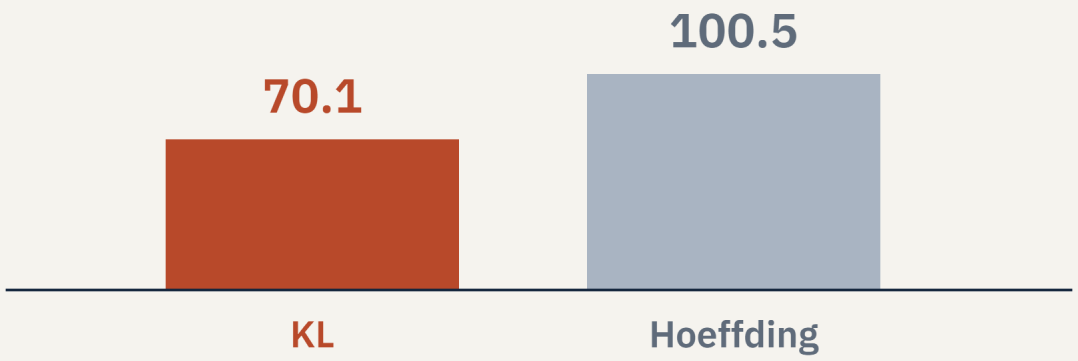
This is a separate ablation, not the baseline comparison of slide 6.

10-trial paired ablation: reuses the main KL runs and runs the matched Hoeffding variant. Source: R1 cells 16–17.

Confidence and cost ablations support separate changes

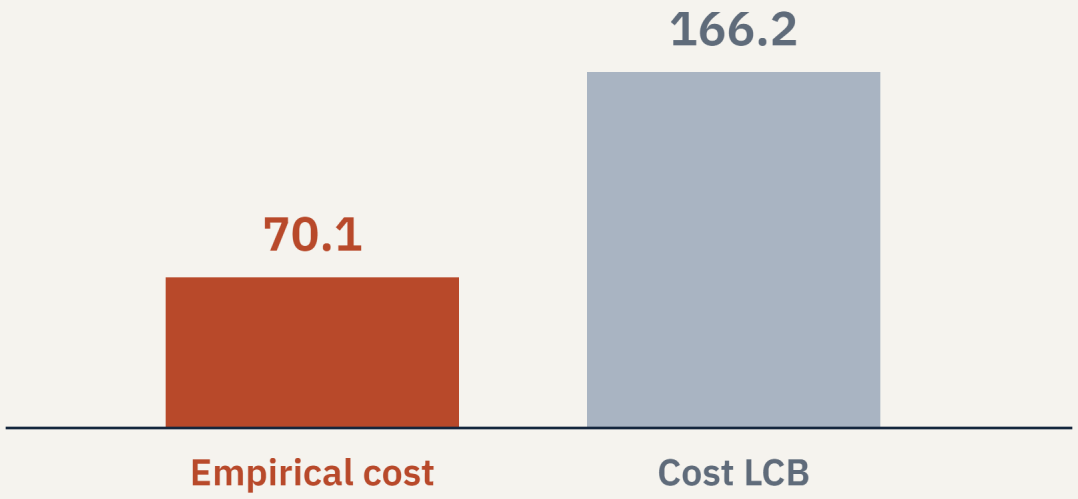
Mean final pseudo-regret · lower is better · same scale in both panels

Confidence bound



Only the confidence bound changes.

Cost estimate in the LP



Mean spend at T/4: 176.0 (empirical) vs 241.4 (cost LCB) of 700.

20 paired trials, one decimal as saved. Each comparison holds the other revised components fixed; neither isolates pacing.

Sources: R2 cell 17 (KL vs Hoeffding), cell 18 (empirical cost vs cost LCB, spend at T/4).

APPENDIX A4 · SCENARIO 3 · ABLATION

The guard ablation shows no measured performance gain

With calibrated rates held fixed, the two guards have identical displayed outcomes over eight paired trials.

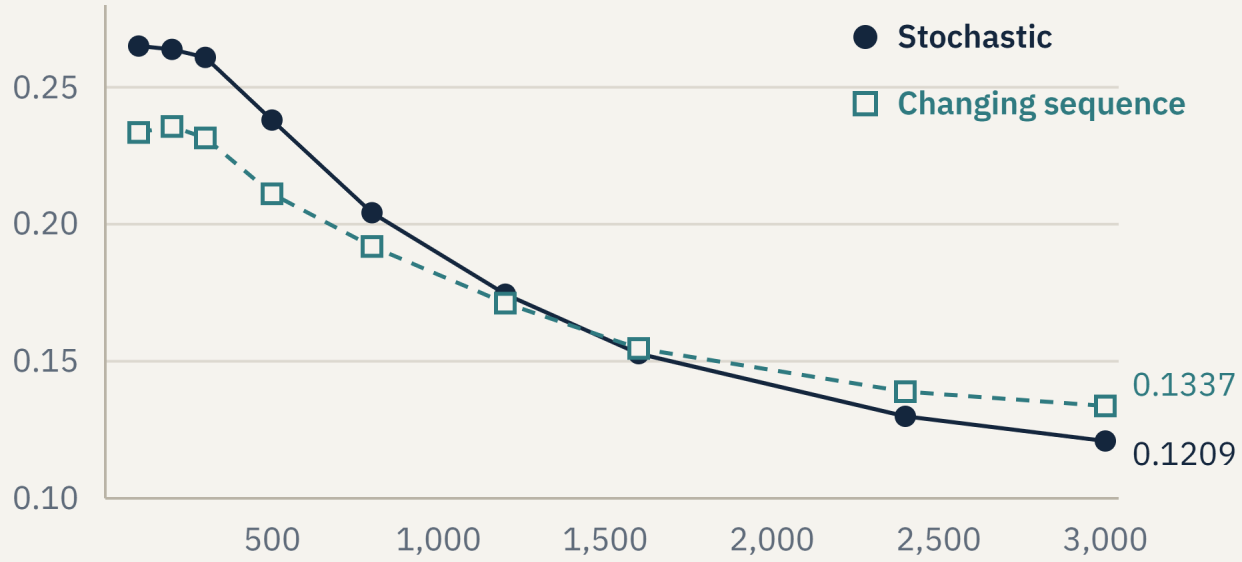
Environment	Guard	Utility	Pseudo-regret / round	Spend / 700
Stochastic	Global stop	1286.3	0.121	674.5
Stochastic	Per action	1286.3	0.121	674.5
Changing sequence	Global stop	1100.3	0.134	597.2
Changing sequence	Per action	1100.3	0.134	597.2

The saved ablation supports rate calibration as the source of the observed improvement on these instances. It does not demonstrate a performance benefit from the guard change.

8-trial ablation means, separate from the 10-trial main means. Source: R3 cells 18–19.

Primal-dual pseudo-regret across tested horizons

Calibrated primal-dual · pseudo-regret per round at each tested horizon



Horizon T (rounds). Markers at the nine tested horizons, 100 to 3,000.

Normalized pseudo-regret decreases overall across the tested horizons, with a small initial increase for the changing sequence.

Eight trials per horizon; B/T held fixed at 700/3,000.

Finite-horizon evidence only: no fitted rate and no asymptotic guarantee.

Source: R3 cell 21 (printed horizon sweep). Comparator: best fixed distribution in hindsight.

Window width and expenditure remain distinct effects

SW-UCB window ablation

Mean final local gap · lower is better · empirical costs and pacing held fixed



Mean spend of 560

Method	Baseline	Revised
SW-UCB	559.98	559.81
CUSUM-UCB	559.97	559.60
Primal-dual	393.41	348.92

PRIMAL-DUAL

Improves utility while spending less, leaving 211.08 of 560 unused.

CUSUM DIAGNOSTIC

Mean resets rise from 18.2 to 32.0; reset counts do not measure detection accuracy.

30 paired trials. Sources: R4 cells 19–20 (window); B4/R4 cell 17 (spend); B4 cell 20, R4 cell 22 (resets).